upSkill CAMPUS

The Education Group
Ghaziabad • Greater Noida
(Estd. : 1995)

I.T.S ENGINEERING COLLEGE
GREATER NOIDA
(A NAAC Accredited Engineering College)

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Industrial Internship Report on

" Automated File Organizer"

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Executive Summary
This report details an industrial internship focused on developing an automated file management system. The project, "Automated File Organizer," was designed to address the common problem of digital clutter by automatically categorizing and moving files into designated folders based on their extensions. The project was completed over a 6-week period, providing hands-on experience with Python-based desktop application development.

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1 Preface

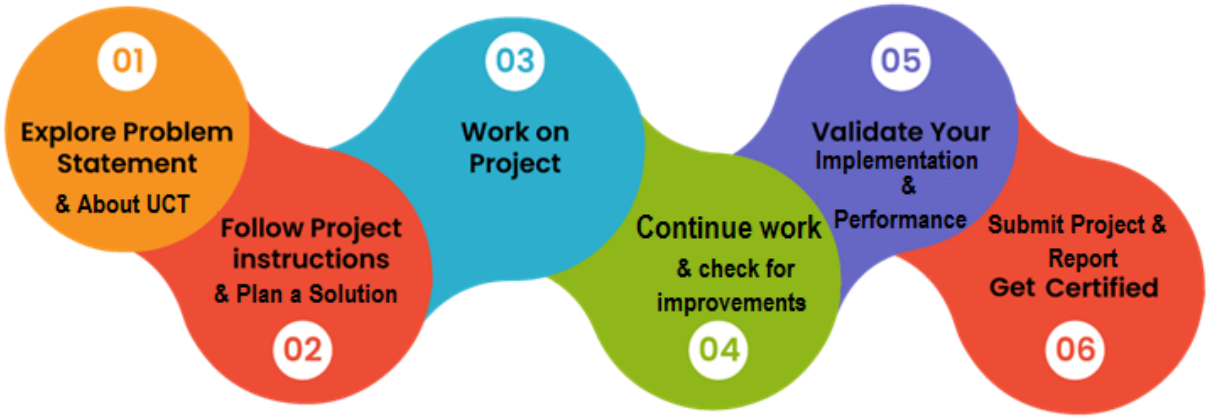
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

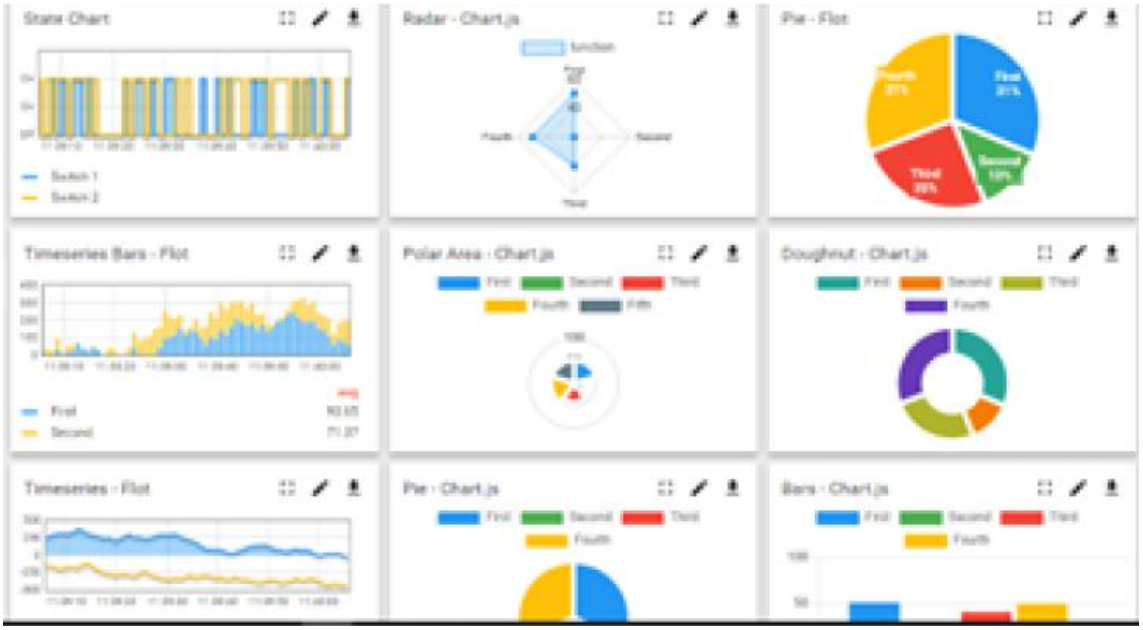
i. UCT IoT Platform (uct Insight)

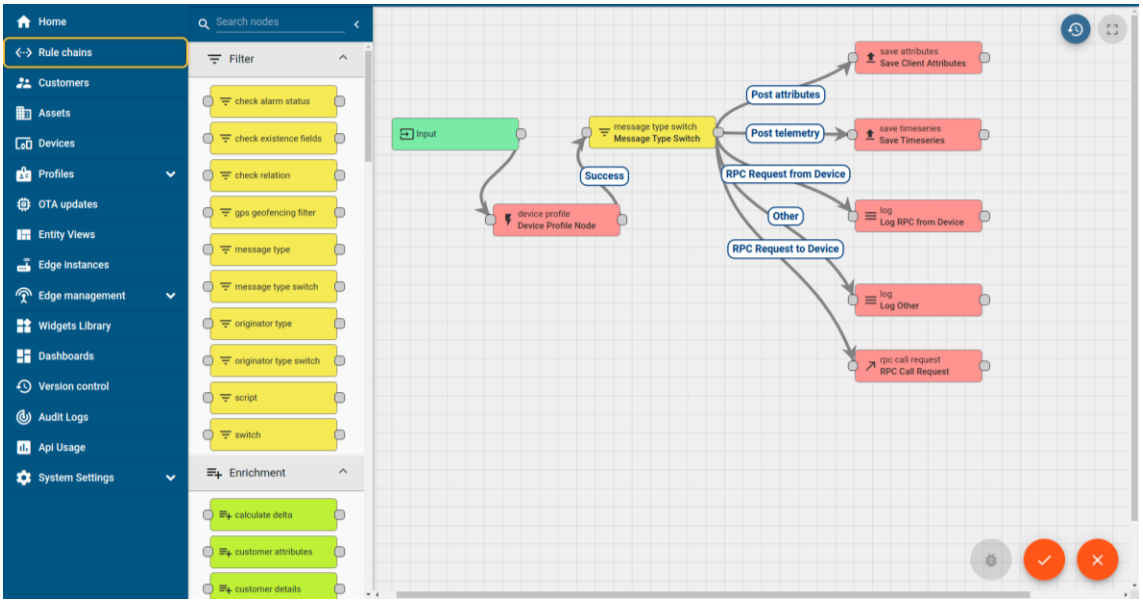
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



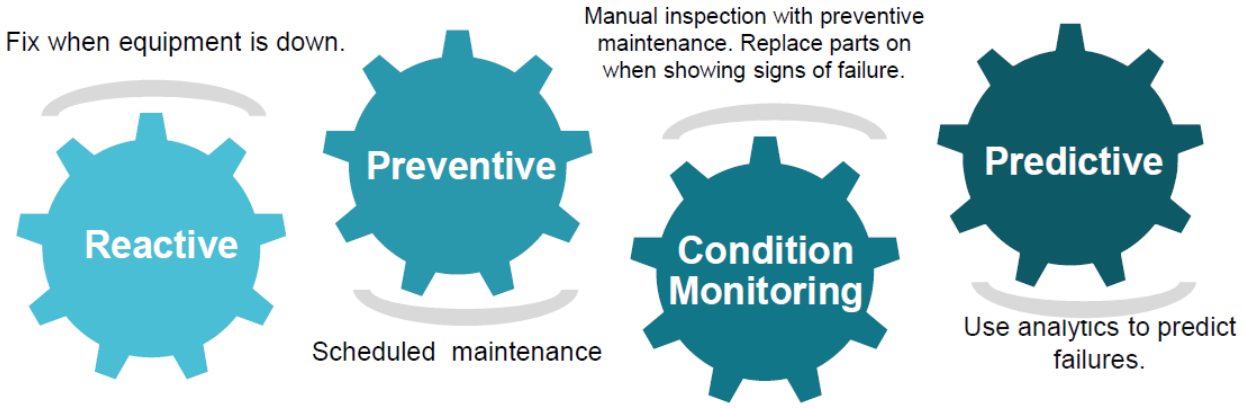


ii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iii. Predictive Maintenance

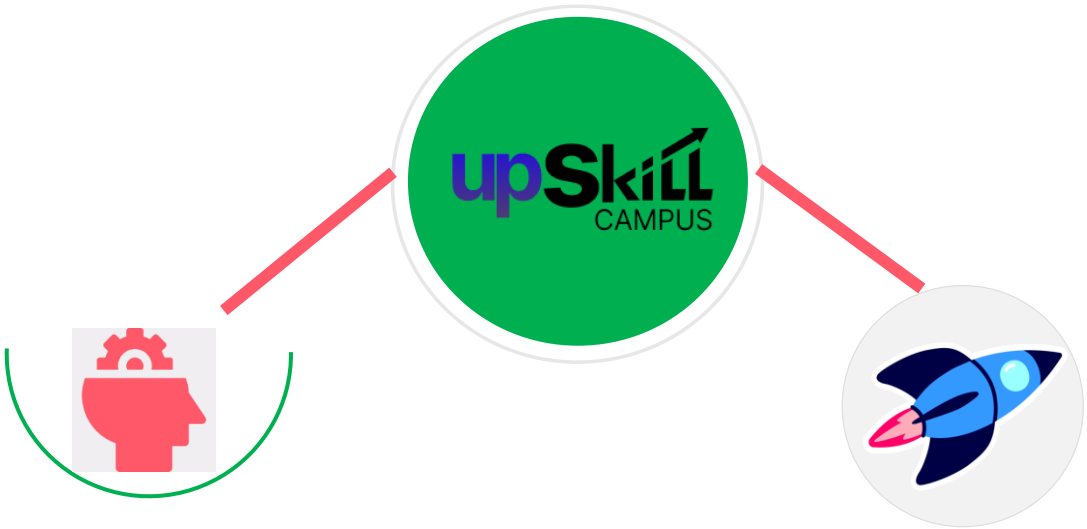
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

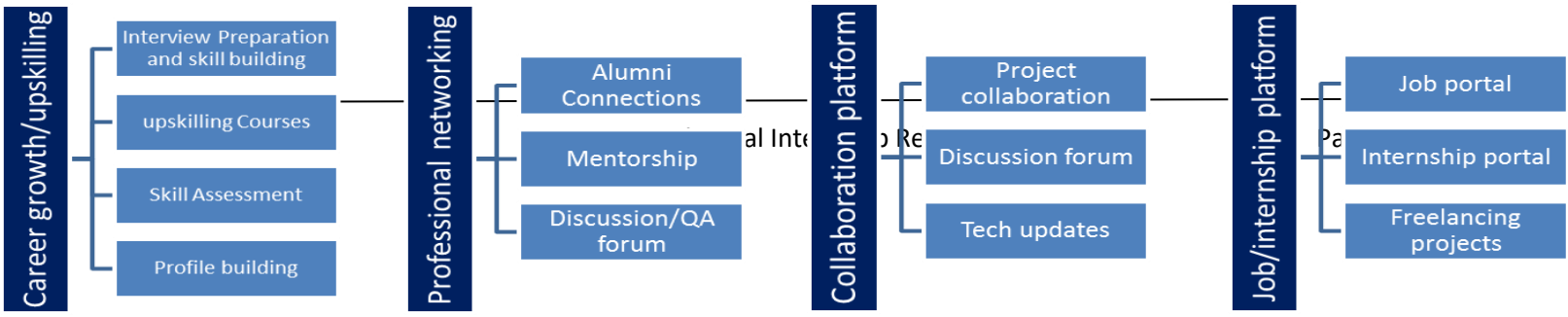
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] UCT Notes

[2] Python Book

[3] Youtube

2.6 Glossary

Terms	Acronym

3 Problem Statement

Users often face difficulty managing large volumes of files in their "Downloads" or "Desktop" folders. Manually sorting these files into categories (Images, Documents, Videos, etc.) is time-consuming and prone to human error.

4 Existing and Proposed solution

Existing Solutions: Manual sorting or complex enterprise-level software that may be too heavy for simple personal use.

Proposed Solution: A lightweight Python application using tkinter for a graphical interface and shutil for file operations.

Value Addition: The tool includes an automated categorization logic that identifies file types (e.g., .jpg, .pdf, .mp3) and organizes them into specific folders like "Images," "Documents," and "Music" with a single click.

4.1 Code submission (Github link)

https://github.com/SHABBIR7033/upskillcampus.git

4.2 Report submission (Github link) :

https://github.com/SHABBIR7033/upskillcampus

5 Proposed Design/ Model

High-Level Design: The system consists of a GUI front-end where the user selects a directory, and a back-end logic engine that scans, categorizes, and moves files.

Logic Flow:

1. User selects a folder path via filedialog.
2. The script retrieves all files using os.listdir().
3. Each file is categorized by its extension (e.g., .docx goes to "Documents").
4. Folders are created if they do not exist, and files are moved using shutil.move().

5.1 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

5.3 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

6 Performance Test

Constraints: The system must handle various file types and ensure that no data is lost during the moving process.

6.1 Test Plan/ Test Cases

Test Cases:

- **Test Case 1:** Organize a folder containing 10 mixed file types (PDF, JPG, MP4).
- **Test Case 2:** Handle a file that has no extension.

6.2 Test Procedure

6.3 Performance Outcome

Outcome: Files were successfully moved to their respective folders without corruption, and the GUI remained responsive during the process.

7 My learnings

During this 6-week internship, I gained significant experience in:

- Developing desktop applications using Python's tkinter library.
- Managing file systems and directory structures programmatically using os and shutil.
- Understanding the importance of industrial-grade code structure and documentation

8 Future work scope

Implement a "Undo" feature to revert file organization.

Add support for more file categories, such as "Source Code" or "Databases".

Integrate a scheduling feature to run the organizer automatically at specific intervals.